How Well Does a Single Mitochondrial Gene (CYTB) Reconstruct *Leopardus* Phylogeny?

There is no universally agreed classification of the genus *Leopardus*, and the number of recognized species varies between approaches used by different groups of scientists. One widely used classification was adopted in 2017 by the IUCN/SSC Cat Specialist Group. It includes 8 species and is based on a combination of morphological, molecular, and biogeographical evidence. The main reference behind it is Kitchener et al. (2017), *A revised taxonomy of the Felidae* [1]. Recently a new species was discovered in the rainforests of Bolivia, which received the name of tilcayo and pretends to be another member of the said genus.

Kingdom: Animalia
└── Phylum: Chordata
  └── Class: Mammalia
    └── Order: Carnivora
      └── Family: Felidae
        └── Subfamily: Felinae
          └── Genus: Leopardus
            ├── Leopardus pardalis — ocelot
            ├── Leopardus tigrinus — northern tiger cat / oncilla
            ├── Leopardus colocola — pampas cat
            ├── Leopardus guigna — guigna / kodkod
            ├── Leopardus wiedii — margay
            ├── Leopardus geoffroyi — Geoffroy's cat
            ├── Leopardus jacobita — Andean mountain cat
            └── Leopardus guttulus — southern tiger cat

Figure 1. Classification of the genus *Leopardus* according to Kitchener et al. (2017), recognizing eight extant species.

This classification lists the species forming the Leopardus (or small cats) as belonging to one genus, however, it does not tell us anything about the evolutionary relations between these species.

A comprehensive study of phylogenetic relations between the species forming the *Leopardus* genus was done by Lescroart et al [2]. In this study, the scientists attempted to reconstruct the phylogeny of the genus using two different approaches. Firstly, they built a maximum-likelihood tree from mitogenomes of individuals representing the species . Secondly, they used WGS data to build separate trees for thousands of genomic regions and then summarize them into consensus trees.

The mitochondrial and nuclear trees coincided in many ways, but were not fully identical. The authors claim that although WGS greatly improves our ability to reconstruct evolutionary history, it does not magically reveal one unquestionably "true tree." Some branches in the consensus tree are supported by not more than 30% of genomic regions. Thus, different regions of the genome tell different stories of evolution.

In this mini-project, we did not aim to resolve the taxonomy of *Leopardus*. Instead, we focused on practicing our skills and used the traditional single-marker approach so to reconstruct a phylogeny based on the mitochondrial **CYTB** gene. Then we compared the result with the published mitogenomic and nuclear genomic relationships.

CYTB was selected because it is a commonly used mitochondrial phylogenetic marker, has an evolutionary rate suitable for distinguishing closely related mammalian species, is approximately 1.1 kb long, and is widely available in public sequence databases.

In our project, we were proceeding in line with the following plan:

1. Download one complete mitogenome for each species + puma → 2. Extract CYTB from each mitogenome→ 3. Align the CYTB sequences → 4. QC the alignment → 5. Select a substitution model → 6. Reconstruct the CYTB tree with bootstrap support → 7. Visualize the reconstructed tree → 8. Compare the tree with the published mitogenomic and WGS consensus trees.

Below, we described each step of the plan in a detailed way.

1. Download one complete mitogenome for each species + puma

For our analysis, we downloaded mitochondrial genomes of the eight *Leopardus* species included in the 2017 IUCN/SSC classification: *L. pardalis, L. tigrinus, L. colocola, L. guigna, L. wiedii, L. geoffroyi, L. jacobita,* and *L. guttulus*. The sequences were obtained from the NCBI RefSeq/GenBank database and downloaded together in GenBank format.

The mitochondrial genome of *Puma concolor* was additionally included as an outgroup. The outgroup is needed to root the phylogenetic tree. The comparisons between the analyzed species with the outgroup species help us determine which patterns in the sequence are more ancestral. Therefore puma gives us an external reference point so we can determine the direction of evolutionary branching and distinguish earlier from later splits within *Leopardus*.

Accession number	Species, Latin	English name	Russian name
NC_085265.1	*Leopardus guttulus*	southern tiger cat / southern oncilla	южная тигровая кошка
NC_028322.1	*Leopardus jacobita*	Andean mountain cat	андская кошка
NC_028321.1	*Leopardus guigna*	guigna / kodkod	кодкод / чилийская кошка
NC_028320.1	*Leopardus geoffroyi*	Geoffroy’s cat	кошка Жоффруа
NC_028318.1	*Leopardus wiedii*	margay	маргай / длиннохвостая кошка
NC_028317.1	*Leopardus tigrinus*	northern tiger cat / oncilla	северная тигровая кошка / онцилла
NC_028315.1	*Leopardus pardalis*	ocelot	оцелот
NC_028314.1	*Leopardus colocola*	pampas cat / colocolo	пампасская кошка
NC_016470.1	*Puma concolor*	puma / cougar	пума

2. Extract CYTB from each mitogenome

We used a short Biopython script to extract the CYTB gene from each mitochondrial genome. The script read the GenBank annotations, searched each record for the coding sequence annotated as **CYTB**, extracted that nucleotide sequence, and saved all nine CYTB sequences into a single FASTA file. The species name and accession number were kept in each FASTA header. All extracted CYTB sequences were 1,140 bp long.

3. Align the CYTB sequences

The extracted CYTB sequences were aligned using **MAFFT** with the `--auto` option, which automatically selected an appropriate alignment strategy, and the resulting alignment was saved as `cytb_aligned.fasta.` MAFFT was chosen because it is a widely used, accurate, and efficient tool for multiple sequence alignment, especially suitable for small sets of closely related DNA sequences.

4. QC the alignment

We performed a quality check of the CYTB alignment to verify the number and length of sequences and to detect possible gaps or ambiguous nucleotides. The alignment contained nine sequences, each 1,140 bp long, with no gaps or ambiguous bases, indicating a complete and consistent alignment of homologous CYTB sequences.

5. Select a substitution model

We used a simple Python script (`alignment_stats.py`) to characterize the CYTB alignment before phylogenetic reconstruction. The script examined each nucleotide position across all nine aligned sequences and classified sites as constant or variable. Variable sites were further divided into parsimony-informative sites, where at least two different nucleotides were each present in at least two sequences, and singleton sites, where the alternative nucleotide occurred in only one sequence. Of the 1,140 positions, 861 were constant and 279 were variable; 145 of the variable sites were parsimony-informative and 134 were singletons.

Site category	Number of sites
Total alignment length	**1,140**
Constant sites	**861**
Variable sites	**279**
Parsimony-informative sites	**145**
Singleton variable sites	**134**

In the next step, we ran **IQ-TREE 2** with **ModelFinder** to compare different nucleotide substitution models and determine which model best described the patterns of sequence variation in our CYTB alignment. The best-fitting model according to BIC was **HKY+F+G4**.

Assumption	Meaning
HKY	The models estimates transitions and transversions in the sequences and allows them to happen at different rates.

Assumption	Meaning
+F	A, C, G and T occur at different frequencies; their observed frequencies are used.
+G4	**Some positions in CYTB change quickly, while others change slowly.** The model puts sites into **4 categories**, from slower to faster.

These assumptions are common in DNA evolution, but they are not equally important for every dataset. In some genes, transitions occur much more often than transversions, while in others the difference is smaller. Likewise, some genes show strong differences in substitution rates between sites, while others are more uniform. Therefore, these features are tested rather than always assumed. For our CYTB alignment, ModelFinder showed that allowing different transition/transversion rates and different evolutionary rates among sites gave a better fit to the data.

6. Reconstruct the CYTB tree with bootstrap support

For phylogenetic tree reconstruction, we used **IQ-TREE 2** selecting the **HKY+F+G4** substitution model. The program searched for the maximum-likelihood tree, i.e. the tree topology that best explains the observed CYTB alignment under this evolutionary model. We used *Puma concolor* as the outgroup to root the tree and performed 1,000 bootstrap replicates to assess the stability of the inferred branches.

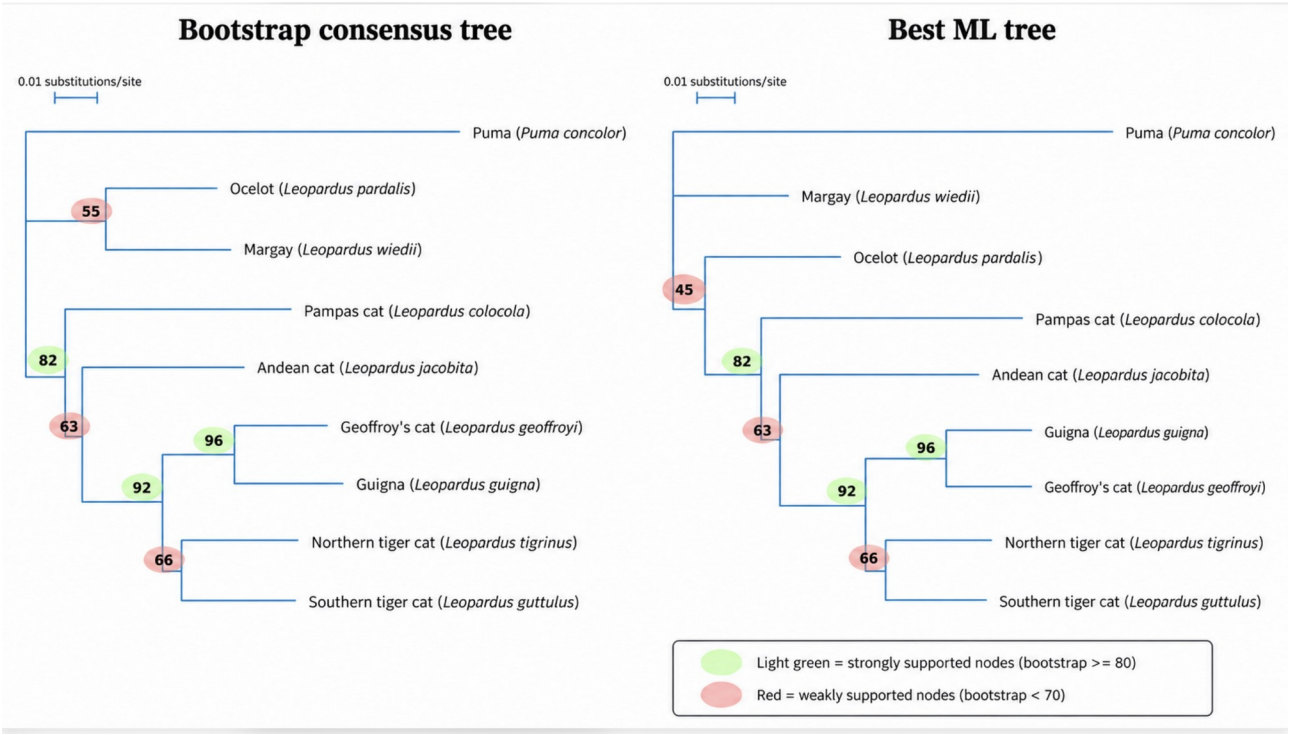
This is the command we used:

```
iqtree2 -s cytb_aligned.fasta -m HKY+F+G4 -b 1000 -o Puma_concolor -pre cytb_ML
```

Bootstrap analysis checks how stable the branches of a phylogenetic tree are. The program repeatedly creates new versions of the alignment by randomly resampling its nucleotide positions. Some positions may be used more than once and some may be left out. A tree is reconstructed from each resampled alignment. If the same branch appears in most of these trees, it receives high bootstrap support, meaning that this relationship is stable.

7. Visualize the reconstructed tree

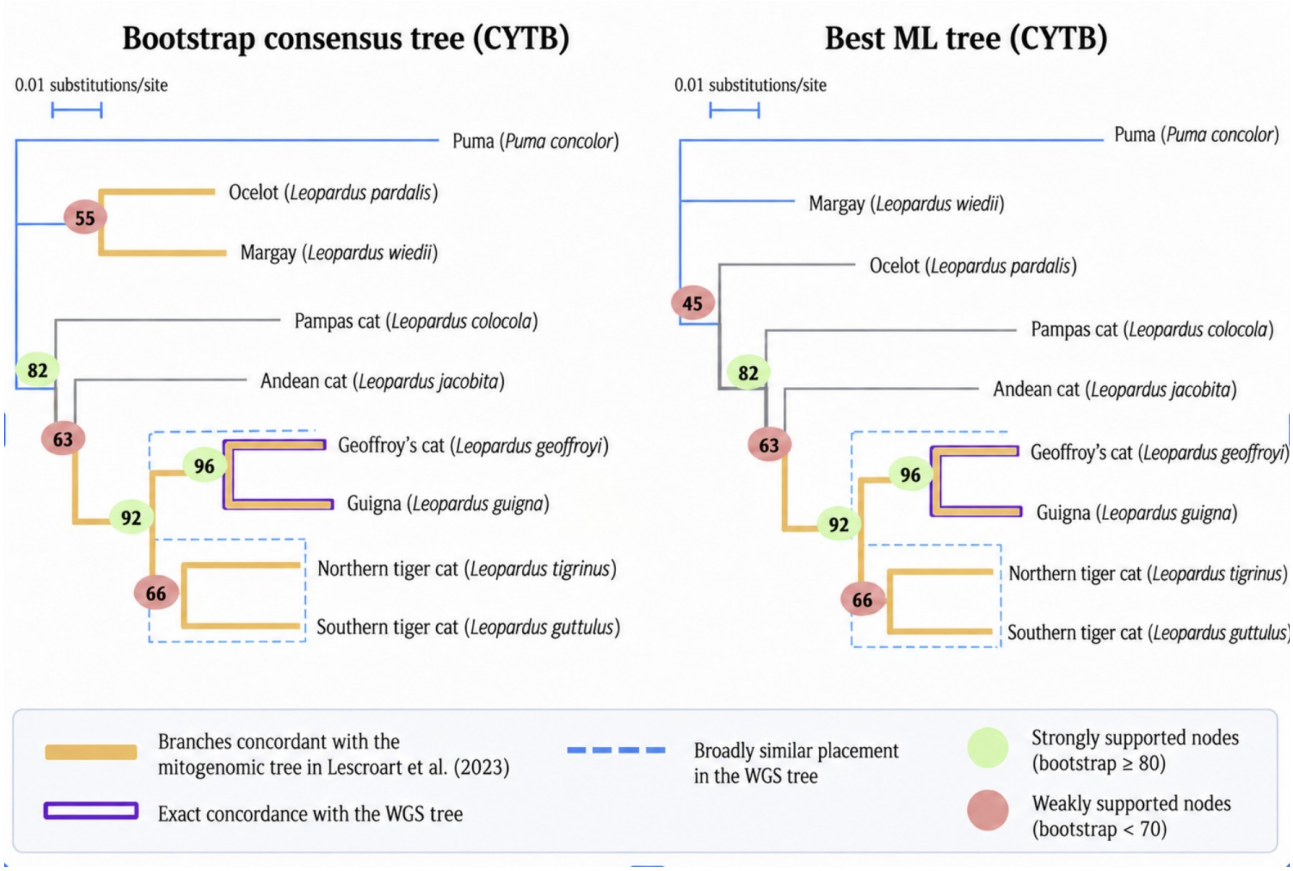
We used iTOL (Interactive Tree Of Life) to visualize the reconstructed phylogenetic tree. The \diagram shows a rooted tree with *Puma concolor* as the outgroup and eight *Leopardus* species forming the ingroup.



Branch lengths represent the estimated number of nucleotide substitutions. For example, the evolutionary distance between *Leopardus guigna* and *Leopardus geoffroyi* is about 0.05 substitutions per site, meaning that on average approximately 5 substitutions occurred per 100 nucleotide sites across the two lineages since their common ancestor. For the 1,140-bp CYTB gene, that corresponds roughly to $0.05 \times 1{,}140 \approx 57$ substitutions.

8. Compare the tree with the published mitogenomic and WGS consensus trees

Our tree partly coincided with the tree reconstructed from complete mitochondrial genomes done by the authors of the article. A sister relationship between Geoffroy's cat and guigna was recovered in both analyses. In the same way as the mitochondrial tree in the article, our phylogeny placed ocelot and margay as sister species. This latter relationship was not supported by the tree built from the whole genome sequences. The Pampas cat and the Andean cat were placed differently in all the three approaches.



Thus, our project shows that a phylogeny based on a single genetic marker can recover important evolutionary relationships, but may be insufficient to resolve all parts of the species history reliably. On the other hand, we need to take into account that the relationships between the species that we considered are highly complicated by such phenomena as incomplete lineage sorting (ILS) and introgression. The authors of the article dive into some of these complications, for instance, they discover mitochondrial capture between the Eastern tiger cat and the pampas cat. The conclusion that we can make from this mini-project is that phylogenetic reconstruction is a complex task. Even if we have very advanced tree-building algorithms and vast data, different genes and genomic regions may support different evolutionary histories, so it is not always possible to recover one simple and unquestionable “true tree.”

References

1. Kitchener, A. C., Breitenmoser-Würsten, C., Eizirik, E., et al. (2017). *A revised taxonomy of the Felidae: The final report of the Cat Classification Task Force of the IUCN/SSC Cat Specialist Group*. *Cat News*, Special Issue 11

2. Jonas Lescroart, Alejandra Bonilla-Sánchez, Constanza Napolitano, Diana L Buitrago-Torres, Héctor E Ramírez-Chaves, Paola Pulido-Santacruz, William J Murphy, Hannes Svardal, Eduardo Eizirik, Extensive Phylogenomic Discordance and the Complex Evolutionary History of the Neotropical Cat Genus *Leopardus*, *Molecular Biology and Evolution*, Volume 40, Issue 12, December 2023, msad255, https://doi.org/10.1093/molbev/msad255